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THE MARMA-CRYPTO PROTOCOL v1.3

Cryptographic-Geometric Watermarking & Sovereign Document Sealing via Sri Yantra Nonlinear Planimetry and Elliptic Curves over $\mathbb{F}_{2^{257}}$

DSTU 4145-2002 · Parallel Multi-Hash Combiner · 256-Bit Bijective Lattice · Canary Leak-Tracing



GEOMETRIC INVARIANT

54 Triple Marmas

9 interlocking triangles form 43 sub-triangles & 54 triple intersections solved to IEEE-754 machine precision (6.13×10^{-11} rad).

BIJECTIVE LATTICE (v1.3)

256-Bit + Canary D5

Encodes full 256-bit signature commitment across 26 coordinates ($|\delta_k| \leq 9.21 \times 10^{-6}$) + Canary Leak-Tracing in `D5.ay`.

ZERO-TRUST VERIFIER

100% Client-Side

Instant verification & suspect leak attribution of `.pdf`, `.docx`, and `.svg` files in your browser or via `apt install`.

Author & Inventor: Yaroslav V. Baienko (@baienko · Attorney-at-Law · Cryptography Researcher)

SWIPE TO EXPLORE →

Why Conventional Watermarks & Detached Signatures Fail

In high-stakes litigation, M&A, and cross-border finance, visual appearance and cryptographic metadata are completely disconnected.

FAILURE MODE A · VISUAL LAYER

Decorative Stamps & Raster Watermarks

- **Zero Cryptographic Binding:** A standard PDF stamp or QR box has no mathematical link to the underlying contract text.
- **Trivial Cloning & Stripping:** Any vector editor can copy a visual stamp onto a forged page or delete a watermark layer in seconds.
- **Raster Degradation:** Pixel-based steganography breaks under vector re-rendering, PDF/A conversion, or scaling.

FAILURE MODE B · METADATA LAYER

Detached CAdES / Hidden ASN.1 Streams

- **Invisible to Human Inspection:** The cryptographic signature lives in hidden binary `/Contents` or external `.p7s` files.
- **Metadata Stripping:** Exporting to `.docx`, `.svg`, or re-saving strips detached containers while leaving the visual page unchanged.
- **Vendor Lock-In:** Requires proprietary national PKI desktop utilities unavailable on headless Linux servers.

THE MARMA-CRYPTO BREAKTHROUGH · DUAL-LAYER CONVERGENCE

Unifying Sacred Nonlinear Geometry with National Elliptic Curve Cryptography

What if the **visible vector geometry on the page** is itself an overdetermined nonlinear mathematical system whose exact vertex coordinates encode the document’s cryptographic signature?

1. VISIBLE VECTOR SEAL

Survives across `.pdf`, `.docx`, and `.svg` vector representations.

2. SUB-MICRON RIGIDITY

Any 10^{-3} coordinate shift breaks 54 triple intersections.

3. STATUTORY QES + TSP

Full PAdES-B-LTA + RFC 3161 TSP court-admissible validity.

Nonlinear Planimetry of the 9-Triangle Sri Yantra

Why the canonical Nava-Cakra is one of the hardest overdetermined problems in computational planimetry.

THEOREM 1 · EUCLIDEAN IMPOSSIBILITY & NONLINEAR CONCURRENCY

9 Intersecting Triangles → 43 Sub-Triangles → 54 Triple Marmas

The central **Nava-Cakra** consists of **4 upward-pointing Śiva triangles** ($\Delta_{U1} \dots \Delta_{U4}$) and **5 downward-pointing Śakti triangles** ($\nabla_{D1} \dots \nabla_{D5}$) inscribed within a canonical circle of radius $R = 100$ centered at $(C_x, C_y) = (150, 150)$.

$$\det \begin{pmatrix} A_{i,1} & B_{i,1} & C_{i,1} \\ A_{i,2} & B_{i,2} & C_{i,2} \\ A_{i,3} & B_{i,3} & C_{i,3} \end{pmatrix} = 0, \quad \forall i \in \{1, ..., 54\}, \quad \Delta E \approx 6.13 \times 10^{-11} \text{ rad} < \varepsilon_{\text{float64}}$$

CANONICAL BASELINE COORDINATES (300 × 300 COORDINATE SPACE)

Exact Bilateral Symmetry Invariant: lx_k + rx_k = 300.000000000

Layer	Orientation	lx_k (Left)	rx_k (Right)	by_k (Base)	ay_k (Apex)
D1	Downward (∇)	53.65669560	246.34330440	123.20508076	250.00000000
U1	Upward (Δ)	52.98401103	247.01598897	174.24660561	50.00000000
U3	Upward (Δ)	98.71823313	201.28176687	220.03828947	123.20508076
U2	Upward (Δ)	78.26467998	221.73532002	197.92315674	78.10499178
D3	Downward (∇)	90.48569229	209.51430771	78.10499178	160.66014976
D2	Downward (∇)	80.98384839	219.01615161	103.12199145	220.03828947
U4	Upward (Δ)	114.94885006	185.05114994	160.66014976	103.12199145
D4	Downward (∇)	116.35142605	183.64857395	134.30757627	197.92315674
D5	Downward (∇)	124.61190803	175.38809197	144.79777263	174.24660561

Cryptographic Binding via National Standard DSTU 4145-2002

From canonical text normalization and parallel multi-hash domain separation to deterministic (r, s) over GF(2^257).

STEP 01 · PARALLEL MULTI-HASH COMBINER (RESISTANT TO SINGLE-PRIMITIVE COLLISION)

SHA-256(GOST 34.311-95 || "::" || Kupyna-256 || "::" || SHA-256)

Document text M is extracted in deterministic order, stripped of dynamic seal headers, normalized via Unicode NFC, and combined in parallel across three independent cryptographic hash families:

DSTU GOST 34.311-95

256-bit national legacy digest for QTSP interoperability.

DSTU 7564:2014

“Kupyna-256” sponge-based national cryptographic hash.

FIPS 180-4 SHA-256

Universal cross-runtime content commitment H_{doc}.

STEP 02 · ELLIPTIC CURVE SIGNATURE OVER GF(2^257)

Polynomial-Basis Curve DSTU_PB_257 & RFC 6979 Determinism

We operate over the binary Galois field $\mathbb{F}_{2^{257}} = \mathbb{F}_2[t]/(t^{257} + t^{12} + 1)$ on the non-supersingular elliptic curve:

$$E(\mathbb{F}_{2^{257}}): \quad y^2 + xy = x^3 + ax^2 + b, \quad a = 0, \quad \deg(f(t)) = 257$$

Using **IETF RFC 6979** deterministic nonce derivation over the Parallel Combiner $C(M)$, the signer’s key produces (r, s) bound into a 256-bit verification digest:

$$V = \text{SHA-256}(r_{\text{hex}} \parallel :: \parallel s_{\text{hex}}) \in \{0,1\}^{256}$$

Full 256-Bit Mantissa Encoding & Canary Leak-Tracing

Bijective integer-nanometer lattice encoding all 256 bits across 26 coordinates + recipient attribution in D5.ay.

MATHEMATICAL TRANSFORMATION · 25 × 10-BIT + 1 × 6-BIT EXACT LATTICE (256 BITS)

Lossless Bijective Inversion from Coordinates to 256-Bit Digest V

All **256 bits** of V are partitioned into 25×10-bit words $w_j \in [0, 1023]$ and one 6-bit word $w_{25} \in [0, 63]$ ($w'_{25} = 16w_{25} + 8$), mapped onto an exact integer-nanometer grid across 26 independent symmetric coordinates (`lx_1..lx_9`, `by_1..by_9`, `ay_1..ay_8`):

$$\delta(w) = (18w - 9207) \times 10^{-9} \in [-9.207 \times 10^{-6}, +9.207 \times 10^{-6}], \quad w = \text{round}\left(\frac{10^9\delta + 9207}{18}\right)$$

BILATERAL SYMMETRY INVARIANT & CANARY LEAK-TRACING (--recipient-id)

Exact Counter-Symmetry + 27th Degree of Freedom (Bindu Apex D5.ay)

Horizontal symmetry $x = 150.0$ is preserved identically ($\Delta_{\text{sym}} = 0$), while the 27th coordinate (`D5.ay`) carries a 10-bit Canary Leak-Tracing beacon $w_{\text{canary}} \in [1, 1023]$:

$$\text{lx}'_k = \text{lx}_k + \delta(w_k), \quad \text{rx}'_k = 300 - \text{lx}'_k, \quad \text{ay}'_9 = \text{ay}_9 + \delta\big(w_{\text{canary}}\big)$$

FULL 256-BIT ENTROPY

Also embeds all 256 bits across 24 symmetric Marma pairs (`11 bits/pair = 264-bit` capacity) with zero visual distortion.

CANARY LEAK-TRACING (v1.3)

Zero-plaintext recipient commitment (`--recipient-id`) identifies the exact leaked M&A / board PDF copy (`trace --suspects`).

Detecting a 0.001 Coordinate Shift or 1-Byte Text Edit

Any modification to document text or vector geometry immediately breaks the 9-triangle verification matrix.

STATE A · AUTHENTIC SEAL

✅ All 9 Layers Verified

`Marma_Crypto_Whitepaper_EN.pdf`

- **Content SHA-256:** Matches embedded seal
- **DSTU 4145-2002** (r, s)**:** Valid over $\mathbb{F}_{2^{257}}$
- **Symmetry Error:** $\Delta_{\text{sym}} = 0.000000000000$
- **Coord Drift:** $\Delta_{\text{coord}} = 0.000000000000$

STATE B · SIMULATED ATTACK (+0.001 ON U2)

❌ Tamper Localized at Layer U2

`Attack: U2.by + 0.001 | U2.lx - 0.001`

- **Public Seal Digest:** Hash mismatch detected
- **Layer U2 Drift:** $\Delta_{\text{coord}} = 0.001000000000$
- **Symmetry Breach:** $\Delta_{\text{sym}} = 0.001000000000$
- **Forensic Verdict: REJECTED (FORGERY)**

EXCERPT FROM LIVE 9-TRIANGLE FORENSIC VERIFICATION MATRIX

Pinpointing Sub-Micron Tampering Across All 36 Coordinates

T_k	lx'_k	rx'_k	by'_k	δ_k	Δ_{coord}	Status
D1	53.65669103	246.34330897	123.20507847	-4.57e-6	0.000000000	OK
U1	52.98401843	247.01598157	174.24660931	+7.40e-6	0.000000000	OK
U3	98.71822977	201.28177023	220.03828779	-3.36e-6	0.000000000	OK
U2	**78.26367665**	**221.73532335**	**197.92415508**	-3.33e-6	**0.001000000**	**TAMPER!**
D3	90.48569103	209.51430897	78.10499115	-1.26e-6	0.000000000	OK
D2	80.98384056	219.01615944	103.12198754	-7.83e-6	0.000000000	OK

Why Marma-Crypto v1.3 Outperforms Classical Signatures

Side-by-side comparison against RSA-3072, ECDSA P-256 / Ed25519, Detached CAdES, and Raster Steganography.

MULTI-DIMENSIONAL BENCHMARK MATRIX · CRYPTOGRAPHY × GEOMETRY × EVIDENTIARY LAW

Comparative Architecture Matrix

Criterion / Layer	RSA-3072 (PKCS#1 v2.2)	ECDSA P-256 / Ed25519	Raster / QR Watermark	🔱 Marma-Crypto v1.3.0 (DSTU 4145 × Sri Yantra)
Algebraic Group	Integer Ring $\mathbb{Z}_n^*$	Prime Field $\mathbb{F}_p$	None (Pixels)	**Binary Field $\mathbb{F}_{2^{257}}$**
Key / Sig Size (128-bit)	`3072 / 3072 bit`	`256 / 512 bit`	`N/A`	**`257 / 512 bit` (Carry-less)**
Digest Primitives	Single (`SHA-256`)	Single (`SHA-256`)	None / CRC	**Parallel (`GOST||Kupyna||SHA256`)**
Nonce RNG Safety	Deterministic / PSS	Vulnerable if RNG weak	`N/A`	**Deterministic `RFC 6979`**
Geometric Capacity	❌ 0 bits (Hidden ASN.1)	❌ 0 bits (Hidden ASN.1)	⚠ Fragile pixels	✅ **Full 256-Bit Bijective Lattice**
Tamper Localization	❌ Binary Pass/Fail	❌ Binary Pass/Fail	⚠ Visual degradation	✅ **Exact Layer $T_1..T_9$ (36 coords)**
Canary Leak-Tracing	❌ None	❌ None	❌ Easily cropped	✅ **`D5.ay` Beacon (`--recipient-id`)**
Archival Legal Force	Requires PAdES	Requires PAdES	❌ Zero legal force	✅ **`PAdES-B-LTA + TSP + OTS`**

ADVANTAGE 01 · COMBINER

Multi-Hash Resilience

Parallel domain-separated combiner remains collision-resistant unless GOST, Kupyna-256, and SHA-256 are broken simultaneously.

ADVANTAGE 02 · BIJECTION

256-Bit Recovery

Recovers the entire 256-bit signature commitment directly from the 9-triangle vector coordinates without metadata reliance.

ADVANTAGE 03 · ATTRIBUTION

Canary Leak-Tracing

Modulates the 27th symmetric coordinate (`D5.ay`) to prove which recipient leaked a confidential M&A or litigation PDF.

Statutory Legal Validity & 25-Year Archival Non-Repudiation

Engineered by a practicing litigation & digital economy attorney to satisfy strict courtroom evidentiary standards.

ETSI EN 319 142-1 · ISO 32000-2

PAdES-B-LTA Archival Container

Embeds full X.509 certificate chains, OCSP responses, and CRL revocation data directly inside the PDF Document Security Store (`/DSS`), guaranteeing offline verifiability for **25+ years** even after CA certificate expiration.

IETF RFC 3161 · ACCREDITED QTSP

Qualified Time-Stamp Token (TSP)

Cryptographically anchors the exact second of signing via accredited Qualified Trust Service Providers (**KNEDP / CZO / monobank / CA Diia**), preventing backdating or key-compromise repudiation.

UKRAINIAN LAW NO. 2155-VIII & eIDAS

Statutory Equivalence to Handwritten Seal

Fully compliant with the Law of Ukraine **“On Electronic Identification and Electronic Trust Services”** and EU **eIDAS Regulation (EU) No 910/2014** (`czo.gov.ua/verify` verified).

DECENTRALIZED PRIOR-ART ANCHORING

OpenTimestamps Bitcoin Proofs (.ots)

Every release manifest and whitepaper SHA-256 digest is anchored into the Bitcoin blockchain via **OpenTimestamps**, providing trustless global proof of existence.

SELF-REFERENTIAL CRYPTOGRAPHIC PROOF · 100% BACKWARD COMPATIBLE (v1.2 + v1.3)

The Whitepapers Are Sealed Using the Exact Protocol They Specify

Signer: Yaroslav Volodymyrovych Baienko · CA JSC Universal Bank (monobank) · TSP: 2026-10-17 12:47 UTC

Both the English (`Marma_Crypto_Whitepaper_EN.pdf`) and Ukrainian (`Marma_Crypto_Whitepaper_UA.pdf`) editions carry live PAdES-B-LTA signatures + embedded 9-triangle Sri Yantra seals verified by an 11-suite automated test harness.

Zero-Trust Online Verifier & Canary Leak-Tracing Inspector

Verify privileged legal contracts (.pdf, .docx, .svg) & trace leaked recipient copies directly in your browser.

ATTORNEY-CLIENT PRIVILEGE BY DESIGN · ZERO SERVER UPLOADS

Pure Client-Side Cryptographic & Geometric Engine

Law firms, banks, and courts cannot upload confidential M&A contracts or litigation filings to third-party cloud verifiers. Our web verifier executes **100% locally inside your browser sandbox**:

MULTI-FORMAT PARSER

Drag-and-drop any signed `.pdf`, `.docx` (OOXML custom properties + paragraphs + tables), or `.svg` file for instant verification.

CANARY LEAK-TRACING PANEL

Enter candidate recipient IDs directly in the browser to identify the exact source of a leaked document via `D5.ay` + SHA-256 commitment.

LIVE ATTACK SIMULATOR

One-click **“⚡ Simulate Attack (0.001)”** demonstrates real-time tamper detection and reset right in the browser.

PRINTABLE COURT PROTOCOL

Generate and print a formal bilingual Cryptographic Verification Certificate for court exhibits with one click.

TRY IT LIVE RIGHT NOW IN YOUR BROWSER

https://yaroslavbaienko.github.io/marma-crypto-protocol/

Pre-loaded with the official English & Ukrainian Whitepapers and Canonical Sri Yantra SVG so you can inspect the 54 triple marmas and test the 1-micron attack simulator immediately.

1-Command Installation & GNOME Desktop Integration

Production-tested packages published in our official GPG-signed Debian/Ubuntu APT repository.

OFFICIAL APT REPOSITORY · https://yaroslavbaienko.github.io/legaltech/

Install Both Sovereign LegalTech Suites in Seconds

```
# 1. Trust the repository GPG key (RSA-4096)
curl -fsSL https://yaroslavbaienko.github.io/legaltech/public.gpg | sudo gpg --dearmor -o /etc/apt/
keyrings/legaltech.gpg

# 2. Register the LegalTech APT source
echo "deb [signed-by=/etc/apt/keyrings/legaltech.gpg] https://yaroslavbaienko.github.io/legaltech
stable main" | sudo tee /etc/apt/sources.list.d/legaltech.list

# 3. Install qes-tools & marma-crypto-watermark
sudo apt update && sudo apt install qes-tools marma-crypto-watermark
```

PACKAGE 1 · AMD64 (v1.0.26)

🔐 qes-tools

- **Standards:** DSTU 4145-2002, GOST 34.311-95, Kupyna-256, PAdES-B-LTA, CAdES-X-Long, ASiC-E, RFC 3161 TSP.
- **Hardware Tokens:** DepositSign, IIT Almaz-1K, Crystal-1K, SecureToken-337.
- **Tools:** `qes-sign`, `qes-verify`, `qes-pdf-court`, `qes-ocr`, plus 18 GNOME Files (Nautilus) actions.

PACKAGE 2 · ALL (v1.3.0-1)

🔱 marma-crypto-watermark

- **Formats:** PDF, DOCX, and SVG 256-bit sealing (`--mode both | watermark | seal` + `--recipient-id`).
- **Desktop GUI:** GTK4 / Libadwaita key selector (`marma-dialog`) & 15-min RAM session cache.
- **CLI:** `marma-watermark sign | verify | trace | protocol | selftest` + Nautilus integration.

Enterprise Integration & Forensic Legal Advisory

Partner with us to deploy mathematical document non-repudiation across banking, legal, M&A, and Web3 workflows.

PACKAGE 01 · AUTOMATED PIPELINES & API

🏛️ Enterprise Document Sealing CI/CD

Turnkey integration of **Marma-Crypto + Qualified Electronic Signatures (DSTU 4145-2002 / eIDAS PAdES-B-LTA + RFC 3161 TSP)** into high-volume banking, law firm, M&A data room, and RWA/Web3 document pipelines via headless Linux CLI, Docker, or on-premise REST/gRPC API.

PACKAGE 02 · INSTITUTIONAL CRYPTOGRAPHIC IDENTITY

🔱 Custom Corporate Geometric Seal

Dedicated organizational geometric-cryptographic seal bound to your corporate **HSM / PKCS11 / QES certificate**, custom white-label browser verification portal, and brand-aligned vector templates for high-value contracts, share certificates, and board resolutions.

PACKAGE 03 · LITIGATION & INTERNATIONAL ARBITRATION

⚖️ Forensic Document Authenticity & Expert Opinion

Independent cryptographic and geometric forensic examination of disputed electronic evidence (`.pdf`, `.docx`, `.svg`, `.p7s`), sub-micron tamper localization across all 36 triangle coordinates, and formal legal/technical expert reports for courts and international arbitration.

CONNECT WITH THE AUTHOR & EXPLORE THE ECOSYSTEM

Yaroslav V. Baienko — Attorney-at-Law & Cryptography Researcher

🌐 Verifier: yaroslavbaienko.github.io/marma-crypto-protocol

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🐧 APT Repo: yaroslavbaienko.github.io/legaltech

𝕏 / Product Hunt: @baienko · p/marma-crypto-protocol